Investigation into Measuring Low Levels of Cohesion of Regolith Simulants

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ABSTRACT

Lunar and Martian regoliths are similar to terrestrial soils in that they appear granular but are expected to contain a small amount of cohesion. As such, cohesion in extraterrestrial regoliths pose challenges for future space operations such as in situ resource utilization (ISRU). As part of preparing for human or robotic missions, it would be necessary to perform geotechnical activities similar to those typical of resource extraction on Earth. ISRU requires a better understanding of mechanical properties of regoliths and to recreate those conditions for physical modeling on Earth. The specific goal of this study was to evaluate two methods—vertical cut and simple direct shear testing—and assess their efficacy in reliably measuring small amounts of cohesion in lunar simulants: JSC-1A and GRC-3. Findings indicate that vertical cut testing cohesion estimates for JSC-1A at relative densities 0%–80% ranged between 0.115 and 0.971 kPa and cohesion estimates for GRC-3 at relative densities 0%–80% ranged between 0.190 and 1.872 kPa. Additionally, results show vertical cut testing provides monotonically increasing cohesion estimates with increased relative density, but simple direct shear testing did not establish any particular trend in cohesion measurements.

INTRODUCTION

A renewed interest in a human mission to the Moon following a 40+ year hiatus (NASA, 2018a), and new initiatives related to asteroids (NASA, 2018b) have stimulated the need for a better understanding of lunar regolith. Understanding the expected surface regoliths (soils) will influence humans' ability to explore or inhabit these extraterrestrial bodies. Therefore, it is important to reliably simulate the expected regolith properties for geotechnical aspects of space exploration such as landing, walking, and investigating potential utilization of lunar resources to develop extraterrestrial habitats (NASA, 2017).

Soil shear strength is often expressed by the following mathematical expression for the Mohr-Coulomb failure criterion:

$$\tau_f = \sigma \tan \phi + c \qquad \textbf{(Eq.1)}$$

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where τ_f is shear stress at failure, σ is normal stress on the failure plane, and ϕ and c are the internal friction angle and cohesion of the soil, respectively. Of the two critical measures of soil shear strength--internal friction angle and cohesion--this study explored the more complex phenomenon of cohesion, which is often neglected in analyzing granular terrestrial soils. In terrestrial soils, this inter-particle attractive force is generally due to electrostatic and electromagnetic forces, cementation, primary valence bonding and adhesion (Mitchell & Soga, 2005). Lunar and Martian regoliths are similar to terrestrial soils in that they appear granular, but are expected to exhibit some cohesion (Carrier, 2005; Demidov et al., 2015). The observation of cohesion present in the granular lunar regolith is illustrated in **Fig.1** by the clear vertical edges of an Apollo 11 astronaut's footprint (NASA, 2019). One hypothesis is that apparent cohesion of lunar regolith is from mechanical interlocking of irregular, angular particles (Carrier, 2005) and is estimated as 0.1-1.0 kPa as interpreted from the Apollo model best estimate (Mitchell et al., 1972d and 1974). Cohesion in extraterrestrial regoliths pose challenges for future space operations such as NASA's In-situ Resource Utilization (ISRU) program (Carrier et al., 1991; Obregon, 2018). Therefore, accurate characterization and simulation of cohesion in lunar simulants could ultimately determine the success of geotechnical designs on the Moon, Mars, and beyond.

Figure 1: Photo of Apollo 11 astronaut Buzz Aldrin's boot print on the lunar surface , NASA Apollo 11 image library AS11-40-5876 (OF300).

Many standardized laboratory and field testing methods can be used to attempt measuring cohesion of soils. These include vane shear (ASTM D4648 / D4648M-16); direct shear (ASTM D3080 / D3080M-11); and triaxial (ASTM D4767-11, ASTM D7181-11, and ASTM D2850-15) testing. Although these ASTM standards are widely used and accepted in both professional practice and research, they each have some drawbacks for use in measuring cohesion of lunar simulants. For example, vane shear testing is considered unreliable for dry granular soils with low levels of cohesion such as lunar soil (Li et al., 2013); direct shear testing forces a horizontal failure plane (Bowles, 1992); and both direct shear and triaxial testing often lack the resolution needed for measuring low levels of cohesion. This study employed the non-standardized vertical cut test and simple direct shear test methods to measure low levels of cohesion in two lunar simulants.

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EXPERIMENTAL METHODS

Materials

The study included lunar simulants JSC-1A and GRC-3. JSC-1A is similar in chemical composition, mineralogy, particle size distribution, particle shape, and engineering properties of the mare basalt encountered during the Apollo 14 mission (He, 2010). JSC-1A is made from glass-rich basaltic ash from the San Francisco volcano field (He, 2010). Lunar simulant GRC-3 was developed as a cheaper alternative to JSC-1A, but it does not match the compositional or textural features of the lunar soil (He, 2010). GRC-3 was initially created for vehicle terramechanics testing (Oravec, 2008). General properties of these lunar simulants are summarized in **Table 1**. This study examined JSC-1A and GRC-3 at 0, 20, 40, 60, and 80% relative densities (D_r).

Table 1. General Material Properties of JSC-1A and GRC-3 (He, 2010; Dewoolkar et al., 2018).

Lunar Simulant	D_{10}^* *(mm)*	D_{30}^* *(mm)*	D_{60}^* *(mm)*	D_r *(max)*	D_r *(min)*	*USCS Classification*	*Internal Friction Angle* ϕ *(degrees)*
JSC-1A	0.017	0.042	0.110	2.040	1.560	SP-SM	34.9-59.6
GRC-3	0.021	0.075	0.210	1.939	1.520	SM	30.4-57.2

*the diameter corresponding to 10, 30, or 60% finer in the particle distribution curve.

Vertical Cut Test

Laboratory vertical cut testing has been used to directly measure cohesion of lunar regolith (Li et al., 2013). In contrast, determination of cohesion from standardized direct shear or triaxial testing requires extrapolation of a regression model fit through the data points. Li et al. (2013) developed their vertical cut test method that mimicked the trench testing procedure used in subsurface explorations conducted on the Moon. Because the vertical cut heights are small, they assumed soil's frictional resistance to be very small resulting in soil's shear strength to be only from cohesion (per **Eq.1**).

In the vertical cut test, soil cohesion is determined based on the "critical height" at which an unsupported vertical slope of a soil fails (Li et al., 2013). It appears that the lower bound theorem can be used to explain the interpretation of the vertical cut test results by Li et al. (2013). The lower bound theorem is a limit analysis solution that considers only soil equilibrium and yield to determine the stress distribution where failure will not occur at a lower load (Chen, 2007). In an element of a failed cohesive soil within the slope, the horizontal or minor principal stress is zero and the vertical or major principal stress is the product of the soil depth and soil unit weight (Atkinson, 1993). If a Mohr circle is drawn for this stress condition (**Fig.2a**), applying the Mohr-Coulomb stress criterion and assuming internal friction angle to be zero yields the relationship as illustrated in **Fig.2** and expressed in **Eq.2** (Li et al., 2013).

where c is the estimated cohesion, γ is the unit weight of soil, and H_c is the critical slope height. The vertical cut device used in this study was developed by Obregon (2018). As illustrated in **Fig.3**, the vertical cut device has internal dimensions 65 cm x 30 cm x 30 cm and consists of segmented and removable inner walls (15 partitions each 2 cm tall) and a thin metal cutting blade. Since the vertical cut box is symmetrically designed, two specimens can be

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prepared simultaneously if desired. Following sample preparation, the top partition of the segmented inner wall can be removed. The metal cutting blade can then be inserted at mid-distance between the exterior wall and inner wall to a depth of 2 mm and back-bladed to pull the simulant away from the vertical edge, allowing excess material to fall into the open cavity between the two segmentally removable inner walls. The metal cutting blade can then be repeatedly inserted 2 mm into the specimen and the material back-bladed off, removing additional segments of the inner wall as needed. "Critical height" is recorded as the height at which the unsupported vertical slope of the soil specimen fails.

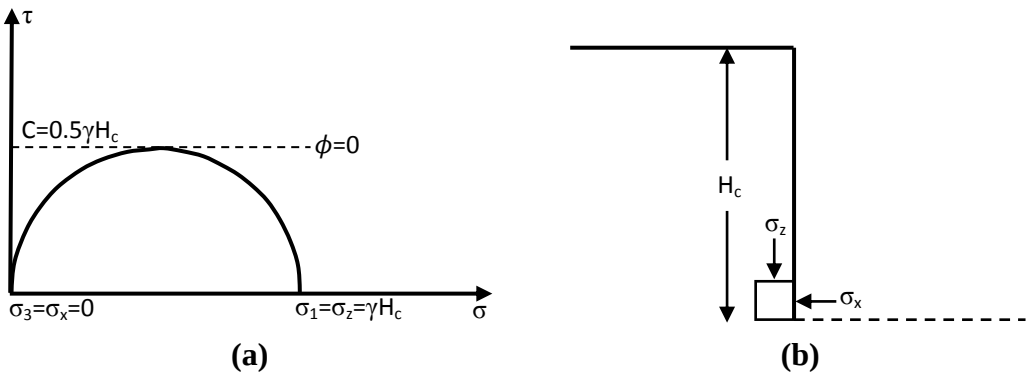

Figure 2: Lower bound solution applied to the interpretation of a vertical cut (a) Mohr circle and (b) stresses on a failed slope element. Where σ_3 or σ_x is the horizontal or minor principal stress, and σ_1 or σ_z is the vertical or major principal stress.

$$c = \frac{1}{2}\gamma H_c \qquad \textbf{(Eq.2)}$$

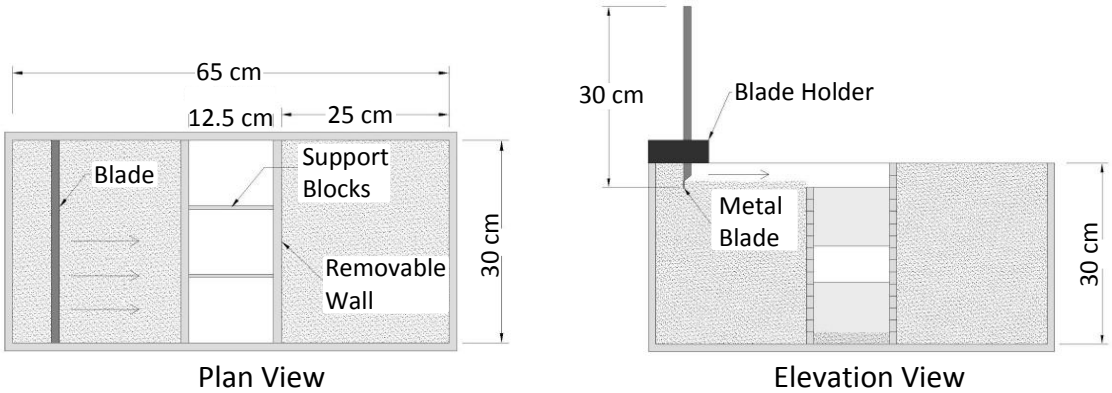

Figure 3: Vertical cut device schematic (Obregon, 2018).

Simple Direct Shear Test

The simple direct shear device used in this study is a simpler version of the direct shear test and allows testing at small normal loads. The device was modeled after the simple direct shear device used by Krantz (1991) and Schellart (2000). This devise was fabricated at the University of Vermont as part of previous research performed by Obregon (2018). As illustrated in **Fig.4**, the simple direct shear apparatus consists of independent upper and lower shear rings (I.D. 8.9

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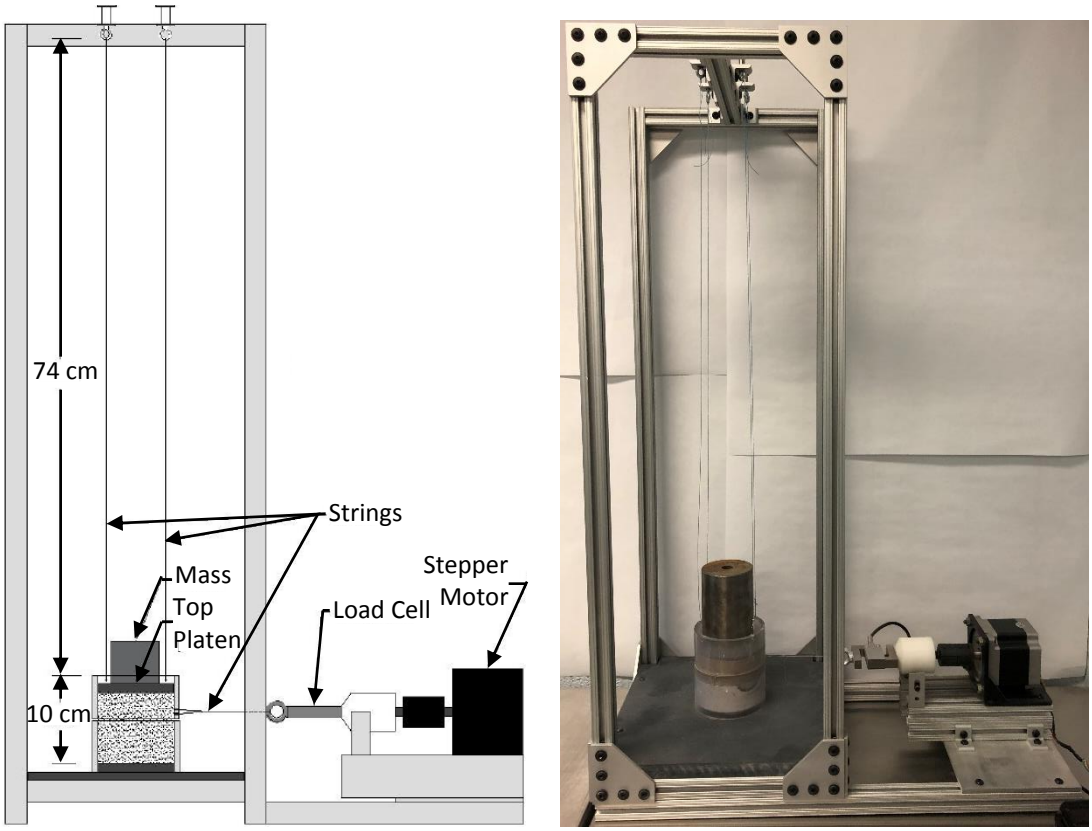
cm dia. x 5 cm) with a fixed bottom ring and suspended top ring connected to a load cell and motor. The top and bottom rings have 6 mm thick walls allowing for a lateral displacement of roughly 5 mm.



(a) **(b)**

Figure 4: Simple direct shear (a) schematic and (b) laboratory test device (Obregon, 2018).

Tests were performed using a sample width-height ratio of 2.5:1, same as standard direct shear test (ASTM D3080 / D3080M-11). Following sample preparation, the specimen was sheared at a rate of 0.01617 mm/s using a Nema 34 stepper motor (StepperOnline, 34HS31-5504S) until peak shear load or 5 mm of displacement, whichever occurred first. Lateral displacements were not allowed beyond the ring wall thickness. Small loads were applied on the top of the specimen that corresponded to applied normal stresses of 0 kPa, 1 kPa, 2 kPa, 5 kPa, and 10 kPa. These loads were held consistent throughout testing. The stress from the weight of soil above the shear plane varied with specimen relative density and was added to the above-mentioned applied stresses giving the total normal stresses (σ) on specimen shear planes. DATAQ Instruments Hardware Manager and WinDaq software (DATAQ Instruments) were used as a data acquisition system to record the load cell voltage readings. The shear stress at failure was computed as the maximum shear load divided by the specimen cross-sectional area:

$$\tau_f = \frac{F}{A_{CS}} \qquad \textbf{(Eq.3)}$$

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where τ_f is shear stress at failure (kPa), F is peak shear force (kN), and A_{cs} is the cross-sectional area of the failure plane (m^2).

Sample Preparation

Although each specimen was considered dry in the "as received" condition, the samples were dried in an oven at 110°C for 24-hours. The second step in sample preparation was to achieve the specified sample densities through uniformly compacted lifts. Relative densities were determined as:

$$D_r = \left[\frac{\rho_d - \rho_{d(min)}}{\rho_{d(max)} - \rho_{d(min)}}\right]\frac{\rho_{d(max)}}{\rho_d} \qquad \textbf{(Eq.4)}$$

where D_r is relative density (%), ρ_d is specimen dry density (kg/m^3), $\rho_{d(min)}$ is the minimum dry density (kg/m^3), and $\rho_{d(max)}$ is the maximum dry density (kg/m^3) (Das, 2010). Published values of minimum and maximum dry densities for GRC-3 and JSC-1A (**Table 1**) were used to calculate the relative densities (ASTM standards D4254-16 and D4253-16; and He, 2010). Specimen densities are summarized in **Table 2**.

Table 2. Dry Densities of Lunar Simulants JSC-1A and GRC-3.

Relative Density (%)	*JSC-1A (g/cm^3)*	*GRC-3 (g/cm^3)*
0	1.560	1.520
20	1.637	1.589
40	1.722	1.664
60	1.816	1.746
80	1.922	1.838
100	2.040	1.939

To ensure the sample density was uniform throughout the entire sample volume, the samples were placed in a series of 2 cm compacted lifts. The mass of sample needed for each lift was calculated prior to placement and compacted using a hand tamp. The vertical cut device required 10 lifts to reach a sample height of 20 cm and the simple direct shear device required one full lift and one partial lift to reach a sample height of 3.56 cm.

RESULTS AND DISCUSSION

Three vertical cut tests were performed at each relative density (0, 20, 40, 60, and 80%) of each lunar simulant for a total of 36 tests. Relative density of 100% was attempted but exceeded the apparatus capacity for both GRC-3 and JSC-1A. **Fig.5** shows a typical (near vertical) failure surface for GRC-3. The critical height test results and average critical height of each specimen series for each relative density are listed in **Table 3**. The three trials performed indicate the repeatability of results.

The results of each specimen test series for each relative density were averaged and the cohesive strength was calculated using **Eq.2**. The cohesive strength of JSC-1A and GRC-3 are

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plotted in **Fig.6**. Li et al. (2013) performed vertical cut tests on JSC-1A and obtained a range of 0.0 to 1.1 kPa for its cohesion for D_r of 13.2 to 86.7 %. The JSC-1A cohesions from the vertical cut tests reported here match those of Li et al. (2013) very well.

Table 3. Vertical cut testing critical heights of JSC-1A and GRC-3.

		Relative Density (%)				
Lunar Simulant		*0*	*20*	*40*	*60*	*80*
JSC-1A (cm)	Trial 1	1.6	2.1	3.0	4.5	10.0
	Trial 2	1.8	1.8	3.7	5.4	10.5
	Trial 3	1.0	2.6	3.4	4.8	9.8
	Average	1.47	2.17	3.37	4.90	10.10
GRC-3 (cm)	Trial 1	2.5	2.4	3.8	8.1	19.8
	Trial 2	2.2	3.2	4.1	8.4	20.4
	Trial 3	2.8	2.8	3.5	9.4	20.9
	Average	2.50	2.80	3.80	8.63	20.37

(a) (b) (c)

Figure 5: Photographs of vertical cut device testing using lunar simulant GRC-3 at 80% relative density (a) cutting blade, (b) mid-test condition, and (c) failure of vertical cut slope.

For a comparative analysis, simple direct shear testing was performed that corresponded to applied normal stresses 0 kPa, 1 kPa, 2 kPa, 5 kPa, and 10 kPa for each lunar simulant at relative densities 0, 20, 40, 60, and 80%. Linear regressions from simple direct shear testing were used to estimate the cohesion (y-intercept) and R^2 correlation coefficient for each relative density of each lunar simulant. All R^2 correlation coefficients were above 0.985, indicating good linear agreement. A comparison of cohesion estimates from vertical cut and simple direct shear testing are presented in **Table 4** and typical simple direct shear test results are shown in **Fig.7**.

Vertical cut testing yielded lower cohesion estimates than simple direct shear testing at relative densities below 60% for JSC-1A or below 40% for GRC-3. GRC-3 data showed greater differences between cohesion values estimated from the two methods. JSC-1A is roughly 36% fines (less than 0.075 mm in diameter) whereas, GRC-3 is roughly 5% fines and GRC-3 is comprised of larger particles as is indicated by the D_{10}, D_{30}, and D_{60} values listed in **Table 1**.

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The differences in the particle distributions and particle shapes may have affected the mechanical interlocking of particles differently causing the observed differences in the cohesion values. Additionally, the simple direct shear testing did not provide a monotonically increasing level of cohesion with increased relative density as would be expected from cohesion derived from the mechanical interlocking of increasingly compact soil. Vertical cut testing, conversely, did provide a monotonically increasing level of cohesion with increased relative density which may indicate it to be a preferred method for estimating low level of cohesion in regolith simulants.

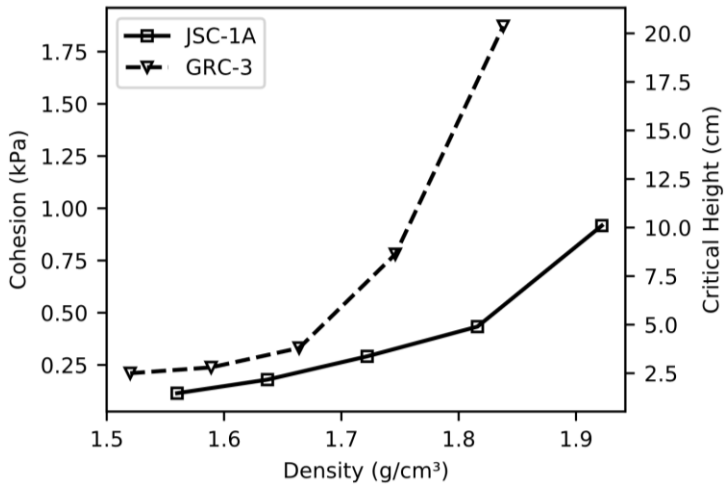

Figure 6: Vertical cut laboratory test cohesion estimates for Lunar simulants JSC-1A and GRC-3.

Table 4. Cohesion estimates of JSC-1A and GRC-3 from vertical cut and simple direct shear testing.

Lunar Simulant	*Test Type*	*Estimate (average)*	*Relative Density (%)* 0	20	40	60	80	100
JSC-1A	Vertical Cut	Cohesion (kPa)	0.115	0.178	0.290	0.445	0.971	N/A
	Simple Direct Shear	Cohesion (kPa)	0.396	0.722	0.413	0.546	0.582	
		R^2	0.987	0.985	0.998	0.996	0.999	
GRC-3	Vertical Cut	Cohesion (kPa)	0.190	0.222	0.316	0.753	1.872	N/A
	Simple Direct Shear	Cohesion (kPa)	0.214	0.282	0.169	0.174	0.463	
		R^2	0.998	0.999	0.999	0.999	0.999	

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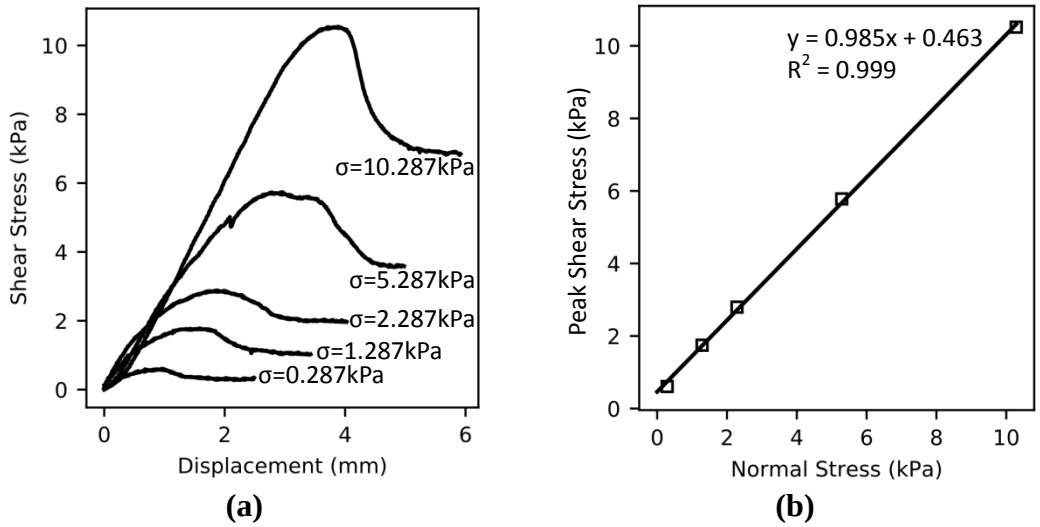

Figure 7: GRC-3 lunar simulant at 80% D_r (a) displacement vs. stress curves for normal stresses 0.287 kPa, 1.287 kPa, 2.287 kPa, 5.287 kPa, and 10.287 kPa at D_r 80%and (b) Mohr-Coulomb failure envelop for the data shown in plot a.

CONCLUSIONS

Cohesion estimates of JSC-1A from vertical cut tests reported here matched well with those obtained by Li et al. (2013) also using vertical cut testing. Additionally, the vertical cut test results as part of this study fall with the range of the Apollo model best estimates. Vertical cut testing-based cohesions of both JSC-1A and GRC-3 increased with increasing relative density. Provided JSC-1A and GRC-3 are suitably representative of lunar regolith, the finding that cohesion is increased with increased relative density aligns with Carrier's hypothesis that apparent cohesion of lunar regolith is from mechanical interlocking of irregular, angular particles. If this hypothesis is correct, the particle size distribution and textural differences between JSC-1A and GRC-3 may explain why the estimated cohesion of GRC-3 was greater than that of JSC-1A when using the vertical cut test method.

Simple direct shear testing of these lunar simulants also provides internal friction angle. The soil internal friction angle in conjunction with the estimated cohesion satisfy the soil properties required to estimate the soil shear strength through the Mohr-Coulomb failure criterion. Cohesion estimates can be, and often are, obtained through regression estimates of the resultant shear-normal curves. Regression models are estimates and this research indicates the need for further work to reconcile the differences between those estimates and the estimates obtained through the use of the vertical cut test method.

Although methods for developing Mohr-Coulomb failure curves such as simple direct shear testing provide shear-normal comparisons required to estimate soil internal friction angle, it may lack the fidelity that is required to reliably estimate low levels of cohesion as are expected in dry granular soil. Through this research, vertical cut testing has shown efficacy as a test procedure to better estimate low levels of cohesion in dry granular terrestrial soils used as lunar regolith simulants if the underlying assumptions made in the interpretation of vertical cut testing are deemed appropriate.

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ACKNOWLEDGMENTS

This work was funded by the Vermont Space Grant Consortium under NASA Cooperative Agreement NNX15AP86H. The authors would also like to thank NASA employees Colin Creager (Mechanical Engineer – Mechanisms and Tribology Branch), Phil Abel (Manager - Mechanisms and Tribology Branch), and Juan Agui (Aerospace Engineer – Glenn Research Center) for their support and guidance. The authors would also like to thank the anonymous reviewers for their constructive feedback.

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